CHAP. 1
INTRODUCTION AND MOTIVATION

References:
→ from textbook: chap. 2 - Statistical Learning

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Text and examples are taken from:
G. James et al., *An Introduction to Statistical Learning: with Applications in R*,
Springer Texts in Statistics, DOI 10.1007/978-1-4614-7138-7 5,

Chapter 1 - content

- 1.1. Overview of Statistical Learning and Machine Learning

- 1.2. Supervised learning and unsupervised learning

- 1.3. Regression problems versus classification problems

Obs.: topics are numbered as they appear in the reference book.

2.1 What Is Statistical Learning?

- Example:
- How to improve sales of a particular product?
  - Advertising data set: sales of that product in 200 different markets
  - advertising budgets for the product in each of those markets for three different media: TV, radio, and newspaper.
  - It is not possible to directly increase sales of the product.
  - On the other hand, we can control the advertising expenditure in each of the three media.
- Goal: to develop an accurate model that can be used to predict sales on the basis of the three media budgets.

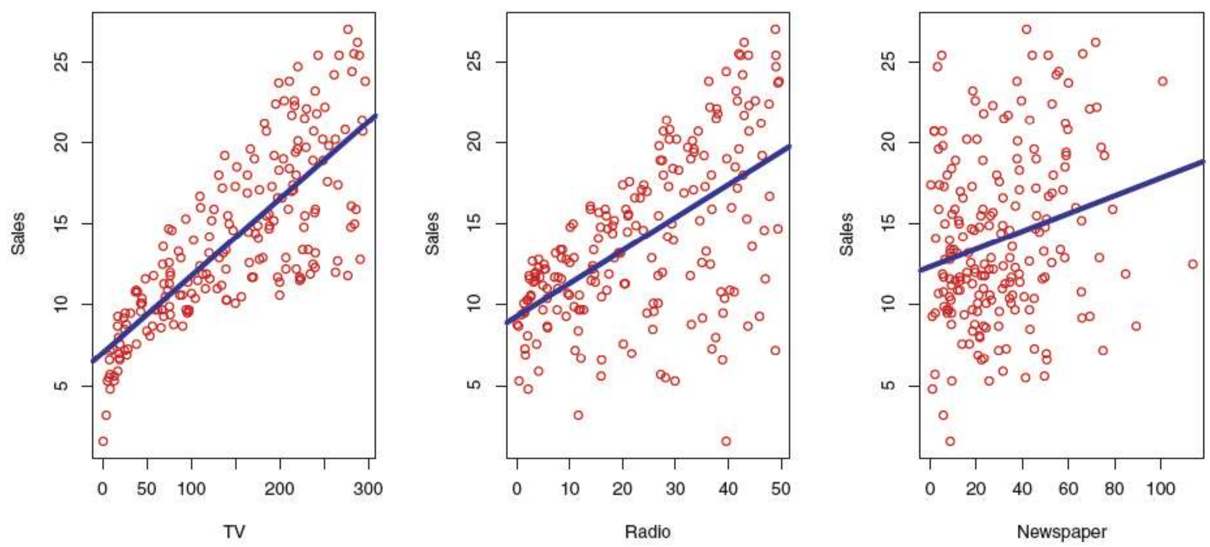

FIGURE 2.1. *The* `Advertising` *data set. The plot displays* `sales`*, in thousands of units, as a function of* `TV`*,* `radio`*, and* `newspaper` *budgets, in thousands of dollars, for 200 different markets. In each plot we show the simple least squares fit of* `sales` *to that variable, as described in Chapter 3. In other words, each blue line represents a simple model that can be used to predict* `sales` *using* `TV`*,* `radio`*, and* `newspaper`*, respectively.*

- advertising budgets: *input variables* → *X* = *(X1, X2, X3)*
  - *predictors*, *independent variables*, *features*, or just *variables*
- Sales is an *output variable (Y)*
  - *Also: response variable or dependent variable*

- General form:

$$Y = f(X) + \epsilon$$

- *f* → some fixed but unknown function of *X*1*, . . . , Xp*,
- And ϵ is a random *error term*, which is independent of *X* and has mean zero

Other examples: f(X1)

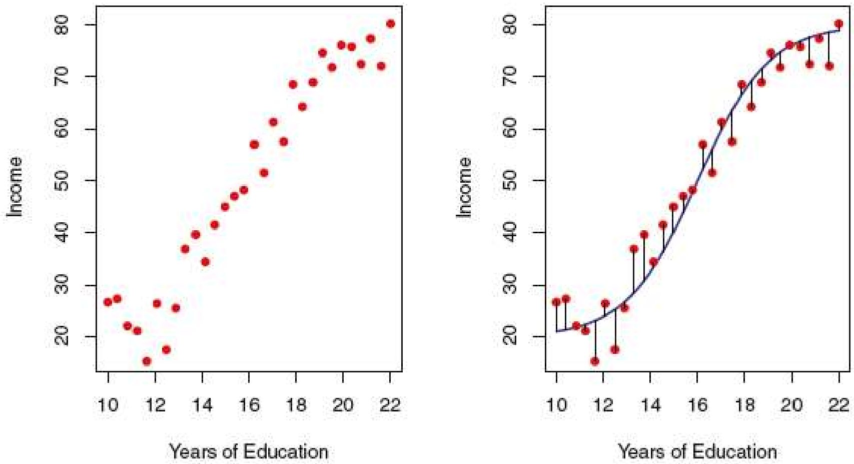

FIGURE 2.2. *The* `Income` *data set.* Left: *The red dots are the observed values of* `income` *(in tens of thousands of dollars) and* `years of education` *for* 30 *individuals.* Right: *The blue curve represents the true underlying relationship between* `income` *and* `years of education`*, which is generally unknown (but is known in this case because the data were simulated). The black lines represent the error associated with each observation. Note that some errors are positive (if an observation lies above the blue curve) and some are negative (if an observation lies below the curve). Overall, these errors have approximately mean zero.*

f(X1,X2)

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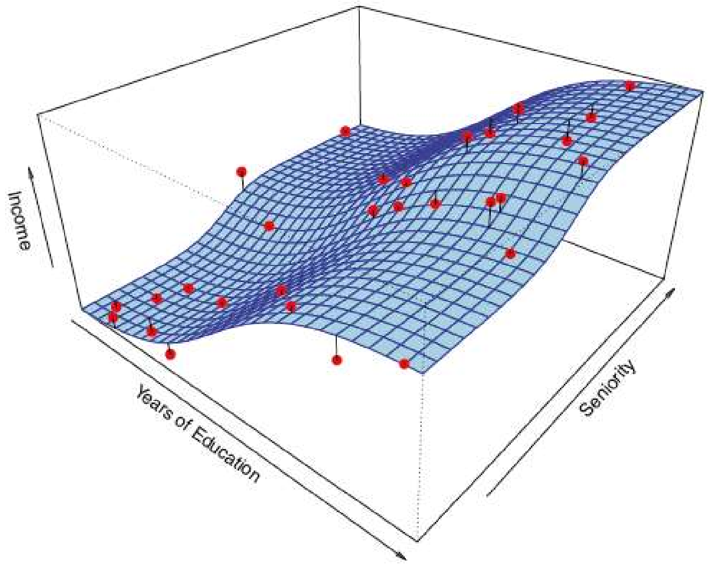


FIGURE 2.3. *The plot displays* income *as a function of* years of education *and* seniority *in the* Income *data set. The blue surface represents the true underlying relationship between* income *and* years of education *and* seniority, *which is known since the data are simulated. The red dots indicate the observed values of these quantities for 30 individuals.*

- In essence, statistical learning refers to a set of approaches for estimating *f*

2.1.1 Why Estimate f ?

Prediction

$$\hat{Y} = \hat{f}(X)$$

- often treated as a *black box*, in the sense that one is not typically concerned with the exact form of $\hat{f}$, provided that it yields accurate predictions for *Y*

$$\begin{aligned} E(Y - \hat{Y})^2 &= E[f(X) + \epsilon - \hat{f}(X)]^2 \\ &= \underbrace{[f(X) - \hat{f}(X)]^2}_{\text{Reducible}} + \underbrace{\text{Var}(\epsilon)}_{\text{Irreducible}} \end{aligned}$$

Inference

- How *Y* is affected as *X*1, . . . , *Xp* change.
- $\hat{f}$ cannot be treated as a black box → know its exact form.

2.1.2 How Do We Estimate *f* ?

- assume that we have observed a set of *n* different data points (*training data*) → estimate *f*.
- apply a statistical learning method to the training data in order to estimate the unknown function *f*.
  - find a function *f* such that $Y \approx \hat{f}(X)$ for any observation (*X, Y*).
- most statistical learning methods for this task can be characterized as either
  - *parametric* or
  - *non-parametric*.

Parametric Methods

- make an assumption about the functional form, or shape, of *f*.
  - use the training data to *fit* or *train* the model
  - If the shape is known, we “only” *have to* estimate a set of parameters.
- potential disadvantage → the model we choose will usually not match the true unknown form of *f*.
  - If the chosen model is too far from the true *f*, then our estimate will be poor.
- one can choose a *flexible* model that can fit many different possible functional forms for *f*.
- However, fitting a more flexible model requires estimating a greater number of parameters.
- More complex models → *overfitting* the data (?)
  - essentially means they follow the errors, or *noise*, too closely.

$$\texttt{income} \approx \beta_0 + \beta_1 \times \texttt{education} + \beta_2 \times \texttt{seniority}$$

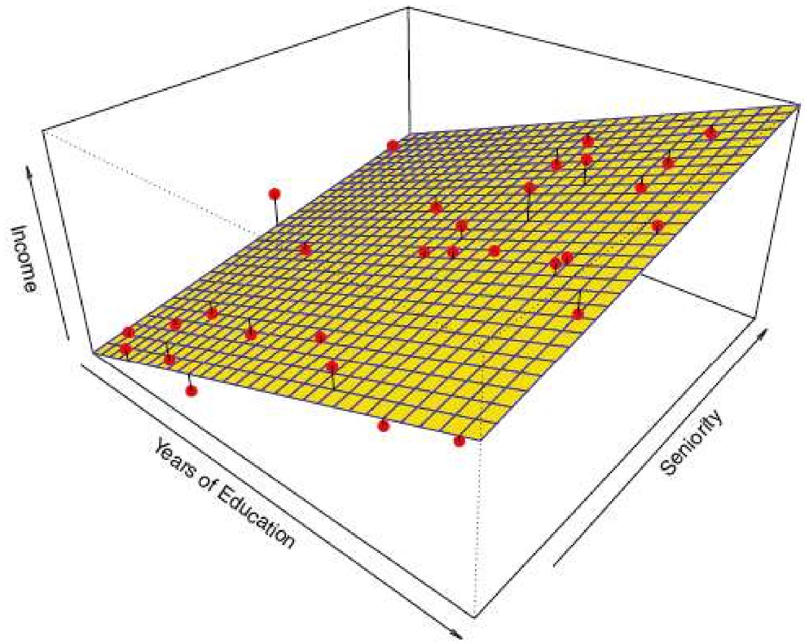


FIGURE 2.4. *A linear model fit by least squares to the* `Income` *data from Figure 2.3. The observations are shown in red, and the yellow plane indicates the least squares fit to the data.*

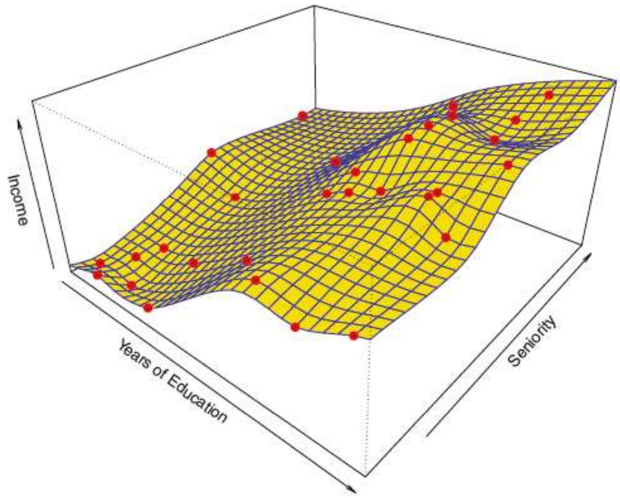
Non-parametric Methods

- do not make assumptions about the functional form of *f*.
  - estimate *f* that gets as close to the data points as possible.

- major advantage over parametric approaches:
  - no assumption of a particular functional form for *f* → potential to accurately fit a wider range of possible shapes for *f*.

- Major disadvantage:
  - do not reduce the problem of estimating *f* to a small number of parameters → requires a very large number of observations (far more than is typically needed for a parametric approach) in order to obtain an accurate estimate for *f*.

Thin-plate spline



A rough thin-plate spline fit to the Income data from Figure 2.3. This fit makes zero errors on the training data.

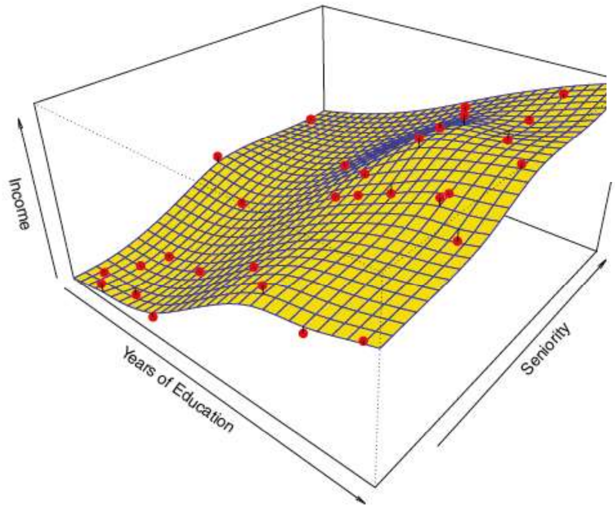


FIGURE 2.5. *A smooth thin-plate spline fit to the* `Income` *data from Figure 2.3 is shown in yellow; the observations are displayed in red. Splines are discussed in Chapter 7.*

2.1.3 The Trade-Off Between Prediction Accuracy and Model Interpretability

- Choice depends on the researcher's interest:

Inference?

Prediction?

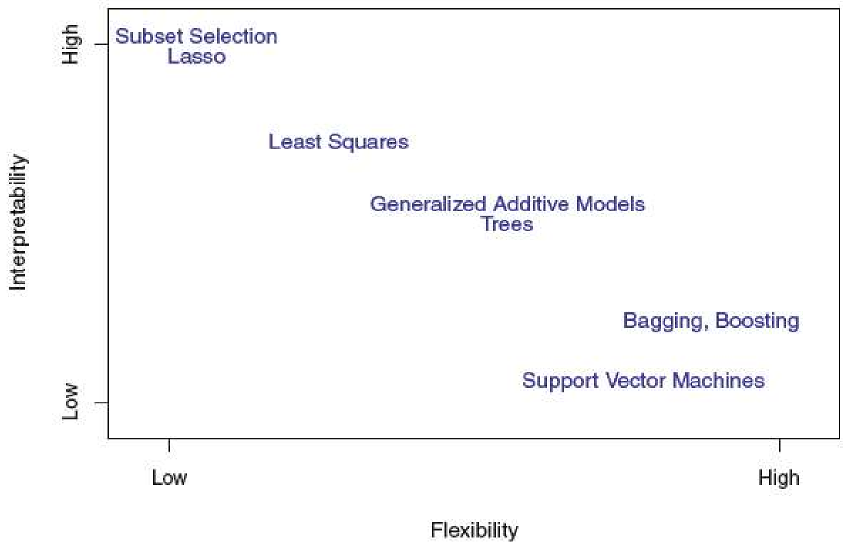


FIGURE 2.7. *A representation of the tradeoff between flexibility and interpretability, using different statistical learning methods. In general, as the flexibility of a method increases, its interpretability decreases.*

2.1.4 Supervised Versus Unsupervised Learning

- Supervised:
  - For each observation of the predictor measurement(s) *xi*, *i* = 1, . . . , *n* there is an associated response measurement *yi*.
  - Wish to fit a model that relates the response to the predictors →
    - Prediction → predicting the response for future observations
    - Inference → better understand the relationship between the response and the predictors.

- Unsupervised:
  - for every observation *i* = 1, . . . , *n*, we observe a vector of measurements *xi* but no associated response *yi*.
  - seek to understand the relationships between the variables or between the observations

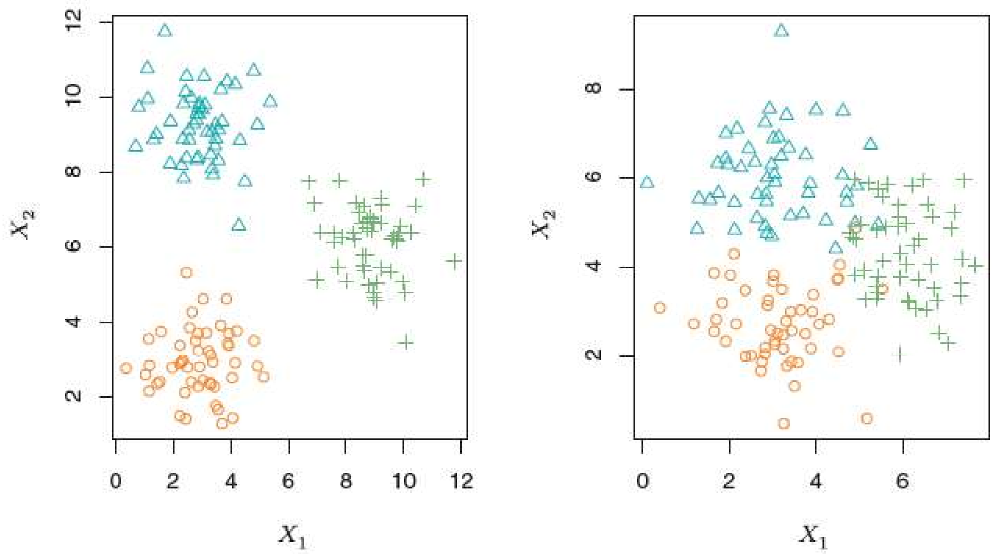


FIGURE 2.8. *A clustering data set involving three groups. Each group is shown using a different colored symbol.* Left: *The three groups are well-separated. In this setting, a clustering approach should successfully identify the three groups.* Right: *There is some overlap among the groups. Now the clustering task is more challenging.*

2.1.5 Regression Versus Classification Problems

- In general:
  - If a quantitative response → *regression* problems,
  - If a qualitative response → *classification* problems.

- *predictors may be* qualitative or quantitative.

- “Most of the statistical learning methods discussed in this book can be applied regardless of the predictor variable type, provided that any qualitative predictors are properly *coded* before the analysis is performed.”

2.2 Assessing Model Accuracy

- different methods may work best for similar (but different) data set.
- for any given set of data which method produces the best results?
- How to select the best approach?
- How to evaluate the performance of a statistical learning method on a given data set?

- we need to quantify the extent to which the predicted response value for a given observation is close to the true response value for that observation. …

2.2.1 Measuring the Quality of Fit

- In the regression setting, the most commonly-used measure is the *mean squared error* (MSE)

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{f}(x_i))^2$$

- Training MSE or Test MSE?

$$\text{Ave}(y_0 - \hat{f}(x_0))^2$$

- select the model for which the average of the test MSE is as small as possible.
- When a given method yields a small training MSE but a large test MSE → we are said to be overfitting the data.

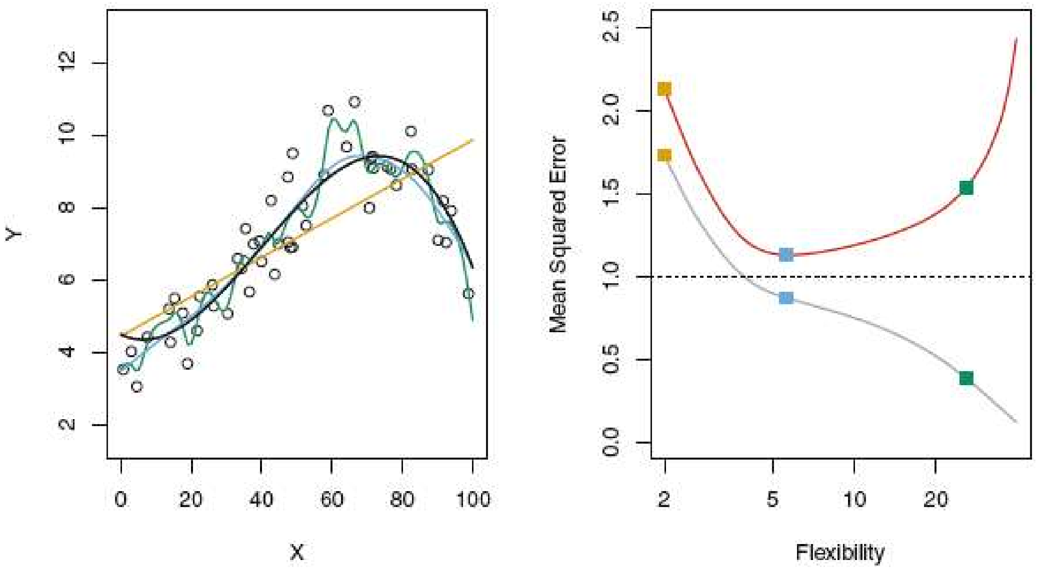


FIGURE 2.9. Left: *Data simulated from f, shown in black. Three estimates of f are shown: the linear regression line (orange curve), and two smoothing spline fits (blue and green curves).* Right: *Training MSE (grey curve), test MSE (red curve), and minimum possible test MSE over all methods (dashed line). Squares represent the training and test MSEs for the three fits shown in the left-hand panel.*

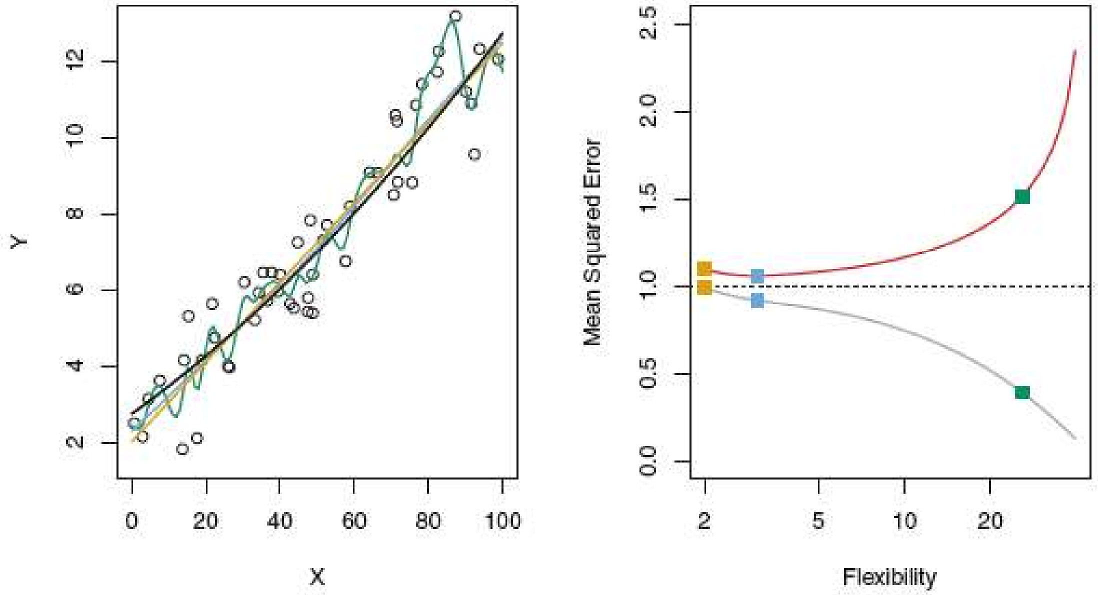


FIGURE 2.10. *Details are as in Figure 2.9, using a different true f that is much closer to linear. In this setting, linear regression provides a very good fit to the data.*

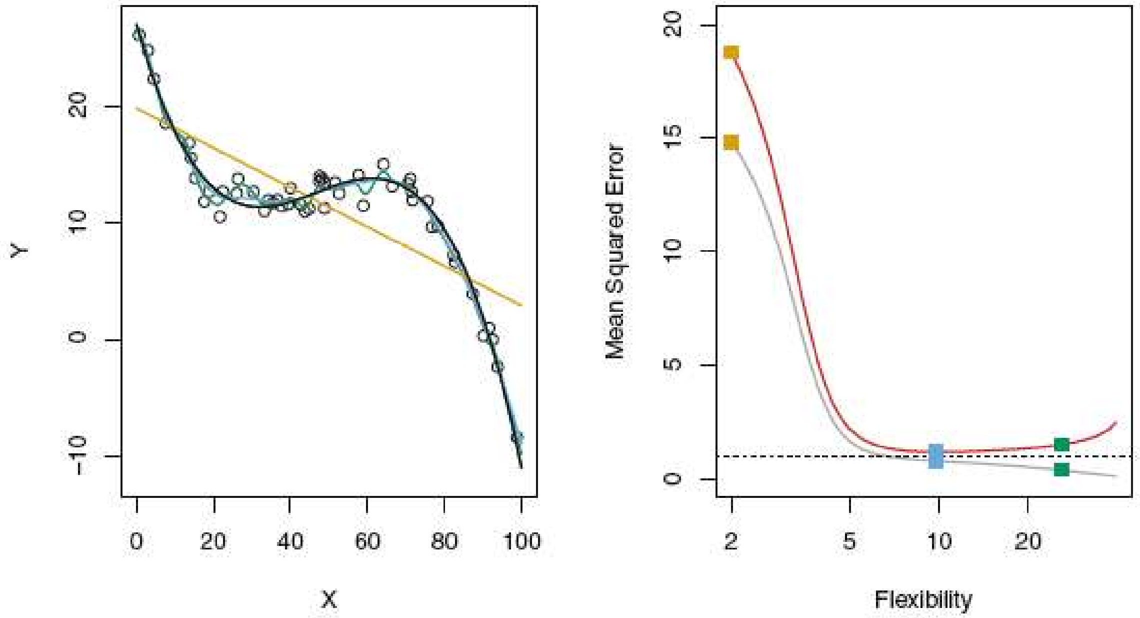


FIGURE 2.11. *Details are as in Figure 2.9, using a different f that is far from linear. In this setting, linear regression provides a very poor fit to the data.*

2.2.2 The Bias-Variance Trade-Off

$$E\left(y_0 - \hat{f}(x_0)\right)^2 = \mathrm{Var}(\hat{f}(x_0)) + [\mathrm{Bias}(\hat{f}(x_0))]^2 + \mathrm{Var}(\epsilon)$$

- *Variance* → amount by which $\hat{f}$ would change if we estimated it using a different training data set.
  - In general, more flexible statistical methods have higher variance.

- *bias* → the error that is introduced by approximating a real-life problem, which may be extremely complicated, by a much simpler model.
  - Generally, more flexible methods result in less bias.

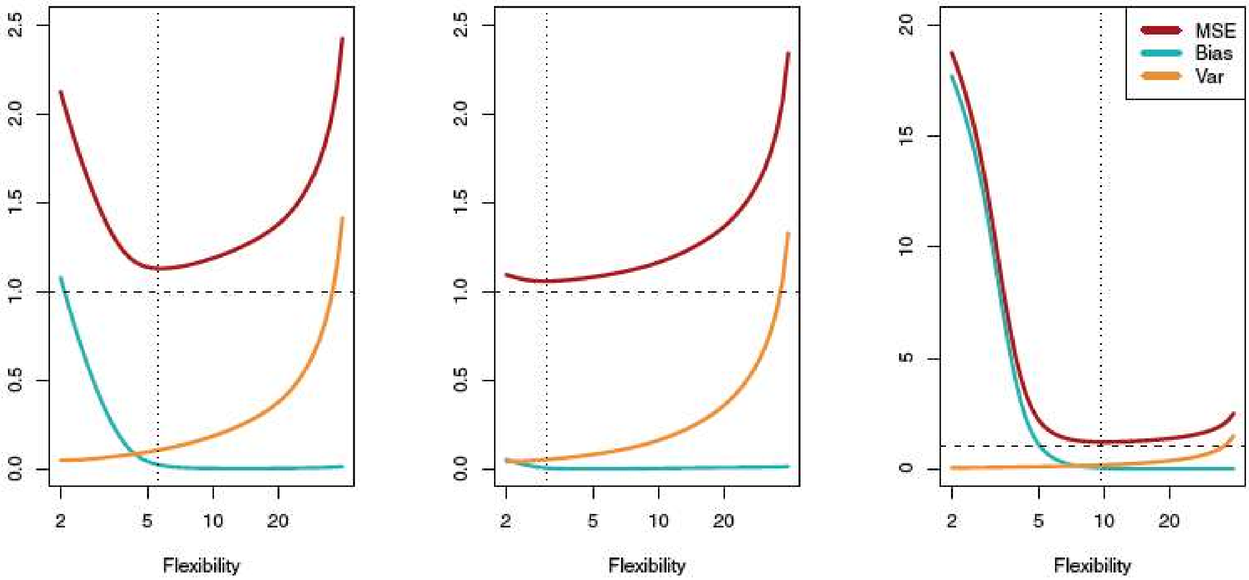


FIGURE 2.12. *Squared bias (blue curve), variance (orange curve), Var(ϵ) (dashed line), and test MSE (red curve) for the three data sets in Figures 2.9–2.11. The vertical dotted line indicates the flexibility level corresponding to the smallest test MSE.*

2.2.3 The Classification Setting

- The most common approach for quantifying the accuracy of our estimate $\hat{f}$ is the training *error rate*
  - the proportion of mistakes that are made if we apply $\hat{f}$ to the training observations:

$$\frac{1}{n}\sum_{i=1}^{n} I(y_i \neq \hat{y}_i)$$

- Training or test *error rate*?

$$\mathrm{Ave}\left(I(y_0 \neq \hat{y}_0)\right)$$

- A *good* classifier is one for which the test *error rate* is smallest.

The Bayes Classifier

- The test error rate is minimized, on average, by a very simple classifier that *assigns each observation to the most likely class, given its predictor values*.
- In other words, we should simply assign a test observation with predictor vector x_0 to the class j for which

$$\Pr(Y = j | X = x_0)$$

is largest.

- overall Bayes error rate: $1 - E\left(\max_j \Pr(Y = j | X)\right)$

- The Bayes error rate is analogous to the irreducible error, discussed earlier.

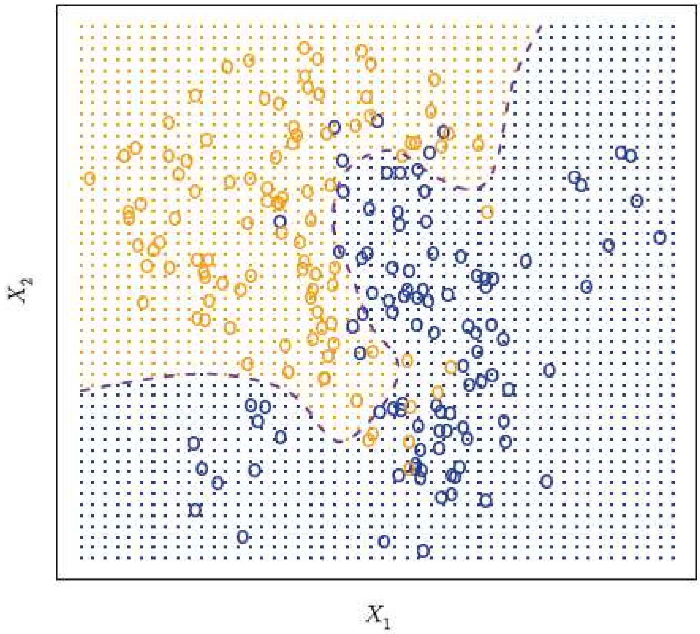


FIGURE 2.13. *A simulated data set consisting of 100 observations in each of two groups, indicated in blue and in orange. The purple dashed line represents the Bayes decision boundary. The orange background grid indicates the region in which a test observation will be assigned to the orange class, and the blue background grid indicates the region in which a test observation will be assigned to the blue class.*

K-Nearest Neighbors

- In theory we would always like to predict qualitative responses using the Bayes classifier.
- But for real data, we do not know the conditional distribution of *Y* given *X*, and so computing the Bayes classifier is impossible.

- KNN → estimate the conditional distribution of *Y* given *X*, and then classify a given observation to the class with highest *estimated* probability

$$\Pr(Y = j | X = x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} I(y_i = j)$$

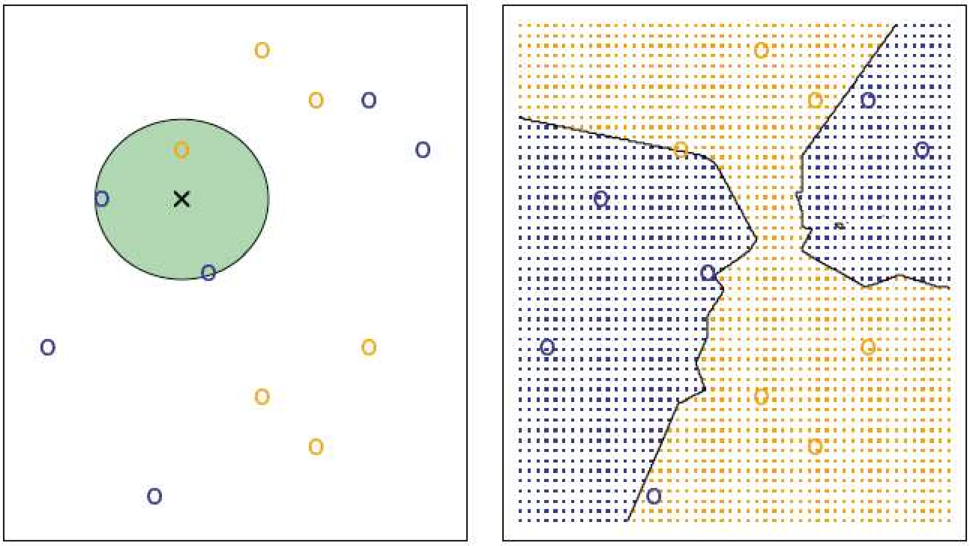


FIGURE 2.14. *The KNN approach, using $K = 3$, is illustrated in a simple situation with six blue observations and six orange observations.* Left: *a test observation at which a predicted class label is desired is shown as a black cross. The three closest points to the test observation are identified, and it is predicted that the test observation belongs to the most commonly-occurring class, in this case blue.* Right: *The KNN decision boundary for this example is shown in black. The blue grid indicates the region in which a test observation will be assigned to the blue class, and the orange grid indicates the region in which it will be assigned to the orange class.*

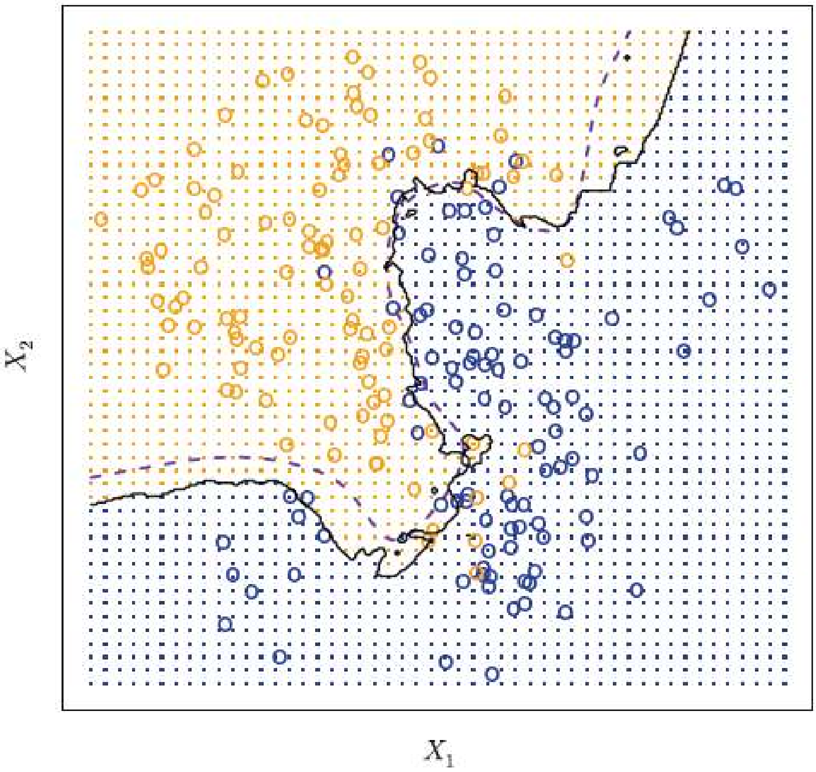


FIGURE 2.15. *The black curve indicates the KNN decision boundary on the data from Figure 2.13, using $K = 10$. The Bayes decision boundary is shown as a purple dashed line. The KNN and Bayes decision boundaries are very similar.*

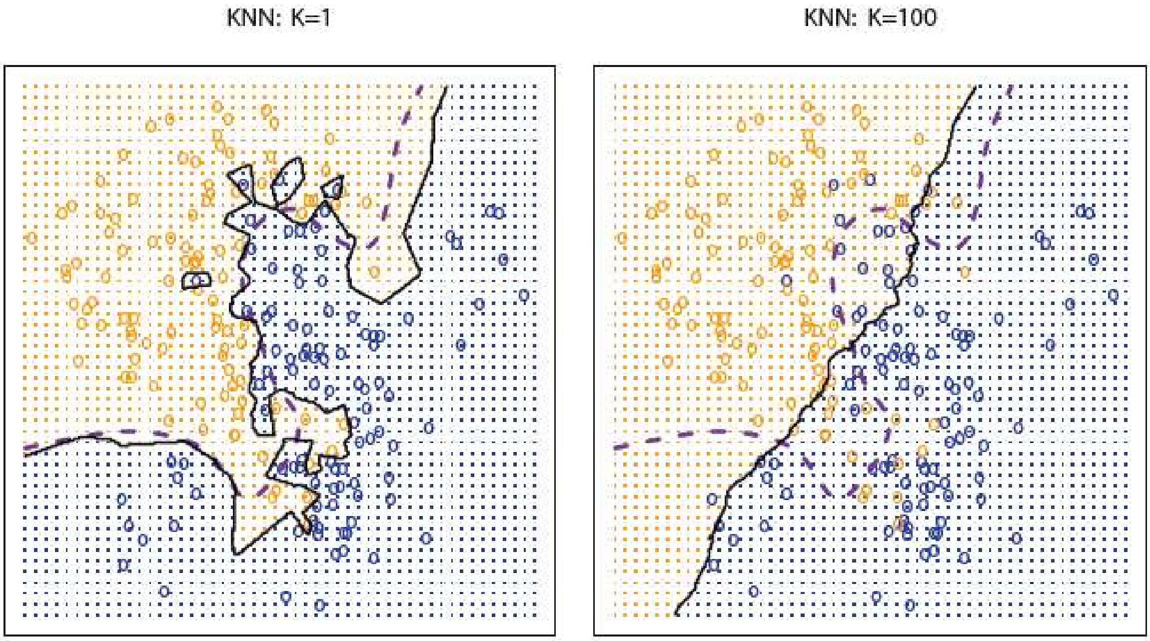


FIGURE 2.16. *A comparison of the KNN decision boundaries (solid black curves) obtained using $K = 1$ and $K = 100$ on the data from Figure 2.13. With $K = 1$, the decision boundary is overly flexible, while with $K = 100$ it is not sufficiently flexible. The Bayes decision boundary is shown as a purple dashed line.*

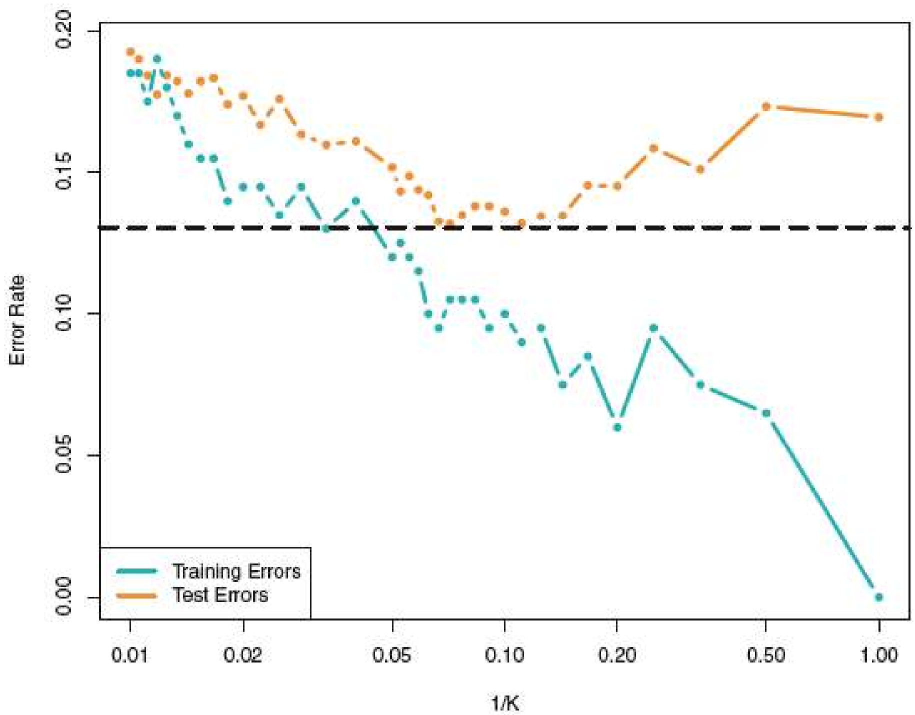


FIGURE 2.17. *The KNN training error rate (blue, 200 observations) and test error rate (orange, 5,000 observations) on the data from Figure 2.13, as the level of flexibility (assessed using $1/K$) increases, or equivalently as the number of neighbors K decreases. The black dashed line indicates the Bayes error rate. The jumpiness of the curves is due to the small size of the training data set.*

2.3 Lab: Introduction to R